

## OFFLINE (Server)

**1 TERA-Gen**  
Synthetic episode generation  
with deterministic onset ( $X, y$ )

**2 Teacher (BiLSTM)**  
Supervised training  
with temporal labels ( $y$ )

**3 Adaptive Distillation (AF-TKD)**  
Knowledge transfer from teacher  
to student via exponential weights

**4 Student (LSTM)**  
Lightweight model obtained via  
adaptive temporal distillation

**5 Model Export**  
Exports LSTM weights (student)  
and metadata to the device

weights of  
LSTM (student)  
+ metadata

## ONLINE (Embedded Device)

**1 Sensor Input**  
Real-time data collection  
(sensors + camera)

**2 MHEG**  
Hybrid gating by entropy and  
motion, frame by frame

**3 LSTM (Student)**  
Lightweight inference only on  
selected frames

**4 Prediction**  
Output  $y(t)$  in real time  
(future event)

### What runs on the embedded device (ONLINE)

- Weight flow: from server to device (once or upon update)
- Sensor Input
  - MHEG
  - LSTM (Student)
  - Prediction (output  $y(t)$ )

### What does NOT run on the device

- TERA-Gen
- Teacher (BiLSTM)
- Adaptive Distillation (AF-TKD)
- Training & Validation